

# Rates and Unit Rates

## Lesson 4-1

Name: \_\_\_\_\_

Date: \_\_\_\_\_

Class: \_\_\_\_\_

## Key Vocabulary

Level 1 support

Picture first, then the word, then a plain-language meaning. Say each word out loud.

*\$12 for 4 games is a rate***Rate**

A ratio comparing two amounts with different units, like miles per hour.

*\$3 per 1 game***Unit Rate**

A rate for just 1 of something, like cost for 1 item.

*60 miles per hour means 60 miles for every 1 hour***Per**

For each one. Example: 5 dollars per book.

*5 to 3 or 5:3***Ratio**

A way to compare two amounts.

*Pack A: \$0.50 per pencil vs Pack B: \$0.40 per pencil — Pack B is the better buy***Better Buy**

The choice that costs less for each item.

 *$\$12 \div 4 \text{ games} = \$3 \text{ per game}$* **Divide**

Splitting a total into equal groups.

## Key Ideas & Notes

- You're comparing ticket prices at different game booths in the arcade.
- Booth A charges \$3 for 5 games, and Booth B charges \$5 for 8 games.
- Which booth gives you a better deal?
- To find out, you need to figure out the cost per game — that's the unit rate!
- Calculate the unit rate (cost per game) for each arcade booth. Divide the total cost by the number of games.

### Think About It

- What two quantities are being compared at each booth?
- Which booth seems like a better deal at first glance?
- How can you compare prices when the number of games is different?

### My Notes

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## Turn & Talk — Discussion Points

Level 1 support

Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

### LAUNCH

**Booth A charges \$3 for 5 games and Booth B charges \$5 for 8 games. Just by looking, which booth seems like the better deal, and what makes it hard to be sure?**

#### Level 1 support

**Start here:** It is hard to compare because the booths have different numbers of games.

**Booth \_\_\_\_ looks better because \_\_\_\_.**

*El puesto \_\_\_\_ se ve mejor porque \_\_\_\_.*

**It is hard to compare because the \_\_\_\_ are different.**

*Es difícil comparar porque los/las \_\_\_\_ son diferentes.*

**Word bank:** rate per ratio compare

**Listen for:** Students notice the prices cannot be compared directly because the number of games differs, so they need the cost for one game (the unit rate) to compare fairly.

#### Level 2

**Why can you not just pick the booth with the lower total price of \$3? Explain what a fair comparison actually needs.**

- You cannot just use the lower price because \_\_\_\_.
- A fair comparison needs \_\_\_\_ so that \_\_\_\_.

EXPLORE

**How did you find the cost per game (the unit rate) for each booth, and what did you divide?**

Level 1 support

**Start here:** I divided the total cost by the number of games to get the cost for one game.

**The unit rate is \_\_\_\_ per one \_\_\_\_.**

*La tasa unitaria es \_\_\_\_ por un \_\_\_\_.*

**I divided \_\_\_\_ by \_\_\_\_ to get the cost per game.**

*Dividí \_\_\_\_ entre \_\_\_\_ para obtener el costo por juego.*

**Word bank:** (unit rate) (divide) (per) (rate) (cost)

**Listen for:** Students explain they divided total cost by number of games (e.g.,  $\$5.00 \div 8 = \$0.625$ ) and state the result as a cost per one game.

Level 2

**Two booths in your table had different totals but the same unit rate. How is that possible, and what does it tell you about the deals?**

- They can have the same unit rate because \_\_\_\_.
- Even though the totals differ, the deals are \_\_\_\_ because \_\_\_\_.

CONNECT

**The snack bar sells popcorn as 3 cups for \$2.25 or 8 cups for \$5.60. How do unit rates help you decide the better buy?**

Level 1 support

**Start here:** I find the price per cup for each bag and pick the lower one.

**The \_\_\_\_ bag is the better buy because its unit rate is \$\_\_\_\_ per cup.**

*La bolsa \_\_\_\_ es la mejor compra porque su tasa unitaria es \$\_\_\_\_ por taza.*

**I divided \_\_\_\_ by \_\_\_\_ to find the price per cup.**

*Dividí \_\_\_\_ entre \_\_\_\_ para hallar el precio por taza.*

**Word bank:** (unit rate) (better buy) (divide) (per) (cost)

**Listen for:** Students compute \$0.75/cup and \$0.70/cup, and choose the large bag because the lower unit rate means each cup costs less.

Level 2

**A friend says the large bag must be the better buy because it costs less per cup AND gives more popcorn. Is 'more popcorn' a fair reason to call it the better buy? Justify using the unit rate.**

- 'More popcorn' is or is not a fair reason because \_\_\_\_.
- The real reason it is the better buy is \_\_\_\_.

## Guided Examples

### Example 1

**A booth charges \$4.50 for 9 games. What is the unit rate?**

**Solution:**  $\$4.50 \div 9 = \$0.50$  per game.

**Answer:** A. \$0.50 per game

### Example 2

**A car travels 180 miles using 6 gallons of gas. What is the unit rate in miles per gallon?**

**Solution:**  $180 \text{ miles} \div 6 \text{ gallons} = 30 \text{ miles per gallon.}$

**Answer:** A. 30 miles per gallon

### Example 3

**A pack of 8 markers costs \$6.00. What is the cost per marker?**

**Solution:**  $\$6.00 \div 8 = \$0.75$  per marker.

**Answer:** A. \$0.75

## Write About the Math

### The Writing Revolution

I can explain my choice using the words rate, unit rate, per, and better buy.

#### 1. Kernel Sentence subject + verb

**Model:** Rate is a ratio comparing two amounts with different units, like miles per hour.

*Tasa es una razón que compara dos cantidades con unidades distintas, como millas por hora.*

**Write a kernel sentence about rate. Use a subject and a verb.**

*Escribe una oración base sobre tasa. Usa un sujeto y un verbo.*

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#### 2. Sentence Expansion because • but • so

**Kernel:** Rate matters in math

*Tasa importa en matemáticas*

Expand the kernel three ways. Add a reason, a contrast, and a result.

**because**  
*porque*

**Rate matters in math because \_\_\_\_.**

*Tasa importa en matemáticas porque \_\_\_\_.*

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**but**  
*pero*

**Rate matters in math, but \_\_\_\_.**

*Tasa importa en matemáticas, pero \_\_\_\_.*

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**so**  
*entonces*

**Rate matters in math, so \_\_\_\_.**

*Tasa importa en matemáticas, entonces \_\_\_\_.*

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### 3. Sentence Types 4 ways to write a math idea

#### Statement *Afirmación*

Tell one true fact about rate.  
*Di un hecho verdadero sobre rate.*

**Rate** \_\_\_\_.

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#### Question *Pregunta*

Ask a question about rate.  
*Haz una pregunta sobre rate.*

**How does** \_\_\_\_ ?

*¿Cómo* \_\_\_\_ ?

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#### Exclamation *Exclamación*

Show excitement about rate.  
*Muestra entusiasmo sobre rate.*

**Wow,** \_\_\_\_ !

*¡Guau,* \_\_\_\_ !

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#### Command *Mandato*

Tell a partner what to do with rate.  
*Dile a un compañero qué hacer con rate.*

**First,** \_\_\_\_ .

*Primero,* \_\_\_\_ .

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### 4. Explain Your Reasoning use a sentence starter

**Booth** \_\_\_\_ **looks better because** \_\_\_\_.

*El puesto* \_\_\_\_ *se ve mejor porque* \_\_\_\_.

**It is hard to compare because the** \_\_\_\_ **are different.**

*Es difícil comparar porque los/las* \_\_\_\_ *son diferentes.*

**The unit rate is** \_\_\_\_ **per one** \_\_\_\_.

*La tasa unitaria es* \_\_\_\_ *por un* \_\_\_\_.

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## Try It

Solve on your own. Check the answer key when you are done.

**1. Which token pack from the table above is the best deal?**

- A. Pack A
- B. Pack B
- C. Pack C
- D. They are all the same

Show your work:

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**2. Runner A completes 4 laps in 12 minutes. Runner B completes 5 laps in 14 minutes. Who runs more laps per minute?**

- A. Runner B
- B. Runner A
- C. Same rate
- D. Cannot determine

Show your work:

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## Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

**An arcade sells tokens in three packs: 25 tokens for \$5, 60 tokens for \$10.80, and 100 tokens for \$19. A student says the 100-token pack is always the best deal because you get the most tokens. Is the student correct? Find all three unit rates and explain.**

*Sentence starter: The student is \_\_\_\_ because the best deal is the \_\_\_\_ pack with a unit rate of \$\_\_\_\_ per token.*

Show your work:

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## Reflect — Exit Ticket

**A store sells 6 pencils for \$1.50 and 10 pencils for \$2.80. Which is the better buy?**

- A. 6 for \$1.50 — \$0.25 each
- B. 10 for \$2.80 — \$0.28 each
- C. They cost the same per pencil
- D. Not enough information

Your answer:

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## Answer Key & Teacher Guide

1. **Try It 1:** B. Pack B — *Pack B has a unit rate of \$0.20 per token, which is the lowest cost per token.*
2. **Try It 2:** A. Runner B — *Runner A:  $4 \div 12 \approx 0.33$  laps/min. Runner B:  $5 \div 14 \approx 0.36$  laps/min. Runner B runs slightly more laps per minute.*
3. **Exit Ticket:** A. 6 for \$1.50 — \$0.25 each —  $\$1.50 \div 6 = \$0.25$  per pencil.  $\$2.80 \div 10 = \$0.28$  per pencil.  $\$0.25 < \$0.28$ , so 6 for \$1.50 is the better buy.

### Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Rate is a ratio comparing two amounts with different units, like miles per hour.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).